Harshana Lakshara Fernando

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, **Portfolio** • LinkedIn • GitHub • Medium

Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka
Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Experience

Resea

Software Engineer Intern *Mar 2026 – Present*
Empathic Computing Laboratory, University of Auckland (Top 100 QS, Remote)

- **KōreroCare: Secure Mobile–Wearable Digital Phenotyping Infrastructure**
  - Designed a research-grade digital phenotyping infrastructure for longitudinal behavioral and physiological data collection across Android, WearOS, and backend services.
  - Engineered reliable passive sensing workflows using Foreground Services and WorkManager to sustain high-uptime data capture under Android OS constraints.
  - Built resilient synchronization, local buffering, and retry mechanisms to reduce data loss and improve consistency across mobile and wearable data streams.
  - Contributed to backend architecture using Go, with service-oriented design for scalable ingestion and secure processing of high-volume time-series data.
  - Integrated Keycloak for authentication and identity management, and introduced Traefik as an API gateway for centralized routing in a containerized environment.
  - Explored Open Wearables integration to support interoperable ingestion and normalization of wearable data across heterogeneous device ecosystems.

KEYWORDS: Kotlin, Android, WearOS, Go, Keycloak, Traefik, Open Wearables, FastAPI, Docker, WorkManager **Software Engineer Intern** *Dec 2024 – May 2025*
Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- **AeroOrder — Revamped Order Management Microservice (Java 17, Spring Boot, Grandle)**
  - Engineered a **distributed**, **high-throughput** unique ID generator for Passenger Name Records (PNRs) using a **Kafka**-backed pool model, ensuring uniqueness and security across scaled instances while adhering to aviation domain standards.
  - Optimized the PNR generation service to achieve **25+ PNRs/sec throughput** per node with **<100ms latency**, validating performance and resilience through extensive load, stress, and spike testing with **Apache JMeter**.
  - Resolved critical **production** issues, including **Kafka** consumer lag and offset resetting, by implementing robust shutdown hooks and enhanced retry mechanisms to improve fault tolerance and system recoverability.
  - Led **Research and Development** for **asynchronous load testing**, establishing a monitoring stack with **Prometheus** and **Grafana** to collect and visualize key performance metrics like queue time and latency.
  - Investigated and refactored inefficient **Kafka Streams** processing logic, building a proof-of-concept to validate a more efficient **Producer-Consumer** model that significantly reduced consumer lag and improved system stability.

- Modernized the logging infrastructure by implementing structured **JSON logging**, which removed legacy dependencies and improved log parsing efficiency across all microservices.

- **AeroMart — Legacy Monolithic Passenger Reservation System (Java 8 ,Ant)**
  - Architected and implemented a hybrid integration strategy by embedding a **gRPC** client and **Kafka** consumer into a **legacy Java 8** monolith (AeroMart), enabling seamless, backward-compatible communication with the new **AeroOrder microservices** platform.
  - Developed dynamic, database-driven **feature toggles** to control PNR routing logic, minimizing risk during the phased **production rollout** by allowing for granular control and immediate fallback.
- **Other**
  - Authored comprehensive **design documents**, **sequence diagrams**, and **integration plans** on Confluence; facilitated knowledge transfer sessions to ensure seamless handover to the core development team.
  - Actively contributed to the **agile development lifecycle**, participating in all ceremonies, performing **code reviews**, and collaborating with cross-functional teams to deliver **production-ready** features using **Jira** and **Bitbucket**.

KEYWORDS: Java 18, Java 8, **Spring Boot**, **Spring Security**, Ant, **gRPC**, **Kafka**, React, Protobuf, REST API, Couchbase, Oracle DB, Oracle SQL, PostgreSQL, **JMeter**, Gatling, Ant Design, **Docker**, Docker Compose, Gradle, **Prometheus**, **Grafana**, Kibana, Confluence, Jira, Bitbucket, Microservices, **Kafka Streams**, JSON

Projects — See more projects on my portfolio

1. Improving Blockchain Scalability: Throughput Enhancement in ZK-Rollups

Research Project — University of Moratuwa

- Built a modular ZK-rollup benchmarking framework to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, proving, data availability, and L1 settlement.
- Implemented the Rust-based Submitter service for finalized batch submission, DA-mode selection, L1 bridge interaction, transaction receipt validation, and settlement metric export.
- Added calldata, EIP-4844-style blob, and off-chain / Validium-style benchmark modes, observing approximately 70% lower DA publication cost for blob-style mode compared with calldata.
- Introduced a mock-testable `BridgeClient` abstraction, improving Submitter testability and increasing verifiable test coverage from approximately 41% to 88%.
- Built Docker-based benchmark orchestration scripts, a Poisson-style workload generator supporting 8 TPS base and 80 TPS burst workloads, and Python data tools for throughput, P50/P95/P99 latency, finalization time, gas/cost, and DA-footprint analysis.
- Produced reproducible benchmark outputs including latency CDFs, cost heatmaps, and Pareto-style trade-off comparisons for the final research evaluation.

Technologies: Rust, Solidity, Hardhat, TypeScript, Python, Docker, gRPC, SQLite/PostgreSQL, RISC0 / Mock Proving Path, Matplotlib

2. Blockchain Supply Chain Management DApp (2025 | Individual) — Architecture | Live | Github

Built a full-stack decentralized application (DApp) enabling end-to-end shipment tracking and secure escrow payments on the Ethereum blockchain (Polygon Amoy Testnet).

- Developed a production-grade Solidity smart contract with a strict 3-state lifecycle (PENDING → IN_TRANSIT → DELIVERED) and on-chain auditability.
- Implemented a trustless escrow mechanism that locks funds on creation and releases payments only upon verified delivery.
- Built a responsive Next.js/React frontend with global state management and Ethers.js integration for secure wallet connections.
- Designed a CI/CD pipeline using GitHub Actions to automate contract and frontend deployment, utilizing Hardhat for local simulation and verification.

Keywords: Solidity, Hardhat, Ethers.js, Web3Modal, Next.js, React Context API, Tailwind CSS, Polygon Amoy, Smart Contracts, Escrow Payments, CI/CD

3. Governance DAO – Token-Weighted Voting DApp (2025 | Individual)

Developed a decentralized governance platform with **token-weighted voting** using Solidity smart contracts.

- Implemented **GovToken** (ERC-20) using OpenZeppelin Contracts, with faucet-based token claims, owner-controlled minting, and airdrops.
- Built **VotingDAO** smart contract enabling proposal creation, weighted voting, quorum enforcement, and post-deadline execution.
- Deployed and tested contracts on Ethereum network using **Hardhat**.

Keywords: Ethereum, Solidity, Web3.js, Hardhat, OpenZeppelin

4. Fintech & Crypto Trading Automation Lab (Ongoing | Group)

Industry-Mentored Initiative (Ongoing)

- Selected for an exclusive initiative to build intelligent automation systems for fintech and crypto markets under industrial mentorship.
- Developing production-ready trading tools and gaining deep exposure to technical analysis (TA) and algorithmic trading strategies.
- Focusing on high-performance software engineering with strict deadlines and client-side expectations.

Keywords: Fintech, Crypto Trading, Forex, Algorithmic Trading, Technical Analysis, Intelligent Automation, Industrial Mentorship, Personal Finance & Wealth Management

Professional Training, Coursework & Certificates

See all certificates

- **Certified Kubernetes Administrator (CKA) Training** — *Apr 2025*
  KodeKloud (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs. View certificate
- **AWS Certified Solutions Architect Associate (SAA-C03) Training** — *Aug 2025*
  Udemy (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course. View certificate
- **Apache Kafka Series (v3) – Learn Kafka for Beginners** — *Mar 2025*
  Udemy (Stephane Maarek) — Full System Design & Setup. [Part 1] | [Part 2] | [Part 3]
- **100 Days of DevOps (Bootcamp)** — *Nov 2025*
  KodeKloud — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible. View certificate
- **Implementing Software Architecture for Large-Scale Systems** — *Apr 2025*
  Udemy View certificate
- **Advanced OAuth Security** — *Mar 2024*
  Udemy View certificate
- **Next.js: Build Scalable React Apps with Page & App Routers** — *Nov 2025*
  Udemy — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support View certificate

Skills

Core	**Java** | **Spring Boot** | **Microservices** | Distributed Systems | System Design | **REST APIs**
Backend	Spring Cloud | Spring Security | Resilience4j | Node.js | Express.js | Gin
Messaging & APIs	**Kafka** | RabbitMQ | **gRPC** | **GraphQL** | WebSockets | Protobuf | JWT | OAuth 2.0
Data	**PostgreSQL** | MySQL | Oracle DB | Redis | Couchbase
Cloud & DevOps	AWS | **Docker** | Kubernetes | CI/CD (GitHub Actions, Jenkins)
Testing	JMeter | Gatling | Pytest | Unit Testing
Tools	Git | Gradle | Postman | Jira | Confluence

References available upon request.